

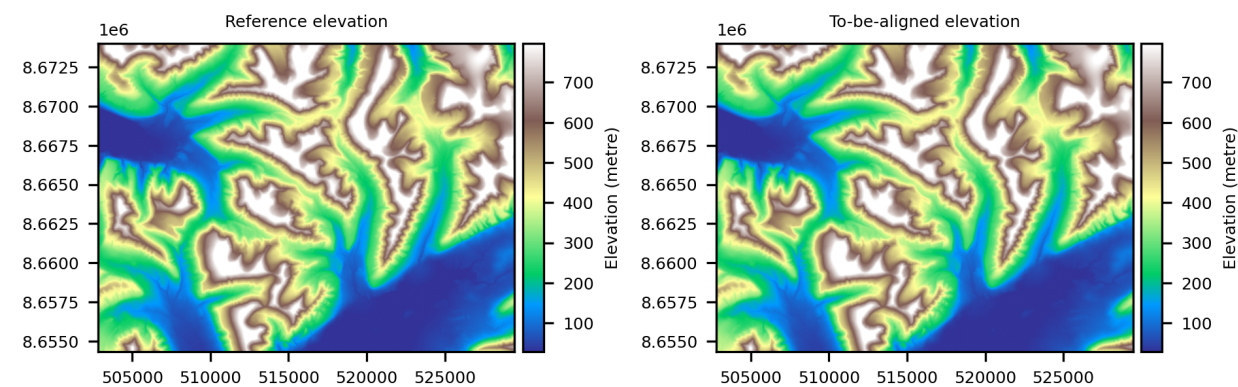
# Accuracy assessment report — xDEM

xDEM version: 0.2.3.dev26

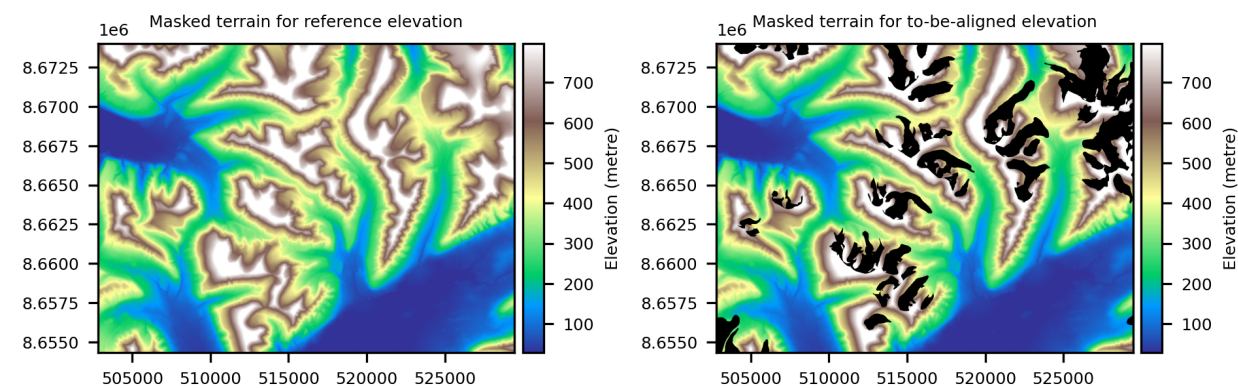
Date: 11/06/2026 16:46:40

Computing time: 17.60 seconds

## Elevation inputs



## Masked elevation data



## Information about inputs

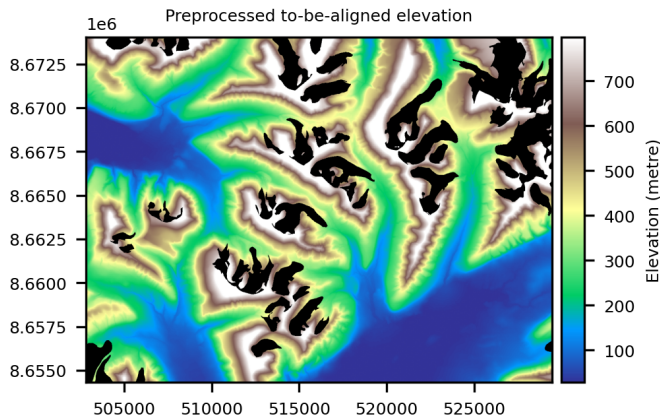
| Information        | Value                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| reference_elev     | {'path_to_elev': '../xdem/example_data/Longyearbyen/data/DEM_2009_ref.tif', 'downsample': '1'}                 |
| to_be_aligned_elev | {'path_to_elev': '../xdem/example_data/Longyearbyen/data/DEM_1990.tif', 'path_to_mask': '../xdem/example_data/ |

```

Longyearbyen/data/glacier_mask/CryoClim_GAO_SJ_1990.shp',
'downsampling': '1'}
sampling_grid      reference_elev

```

## Preprocessed elevation data



## Coregistration user configuration

```

Information Value
step_one      {'method': 'LZD', 'extra_information': '{"subsample': 10000}"}
process       True

```

## LZD inputs

```

Information Value
random        {'subsample': '10000', 'random_state': '42'}
fitorbin      {'fit_minimizer': 'least_squares', 'fit_loss_func': 'linear'}
iterative     {'max_iterations': '200', 'tolerance': '0.01'}
specific      {}
affine        {'only_translation': 'False'}

```

## LZD outputs

```

Information Value
affine        {'centroid': '(515650.83, 8663888.17, 353.11)', 'matrix':
               '[[ 9.99999985e-01 1.17578419e-04 -1.30099078e-04
1.02548967e+01]\n [-1.17594385e-04 9.99999986e-01 -1.22719549e-04

```

```

1.90590312e+00]\n [ 1.30084647e-04 1.22734846e-04 9.99999984e-01
-1.93544013e+00]\n [ 0.00000000e+00 0.00000000e+00
0.00000000e+00 1.00000000e+00]]', 'shift_x': '10.25', 'shift_y': '1.91',
'shift_z': '-1.94'}

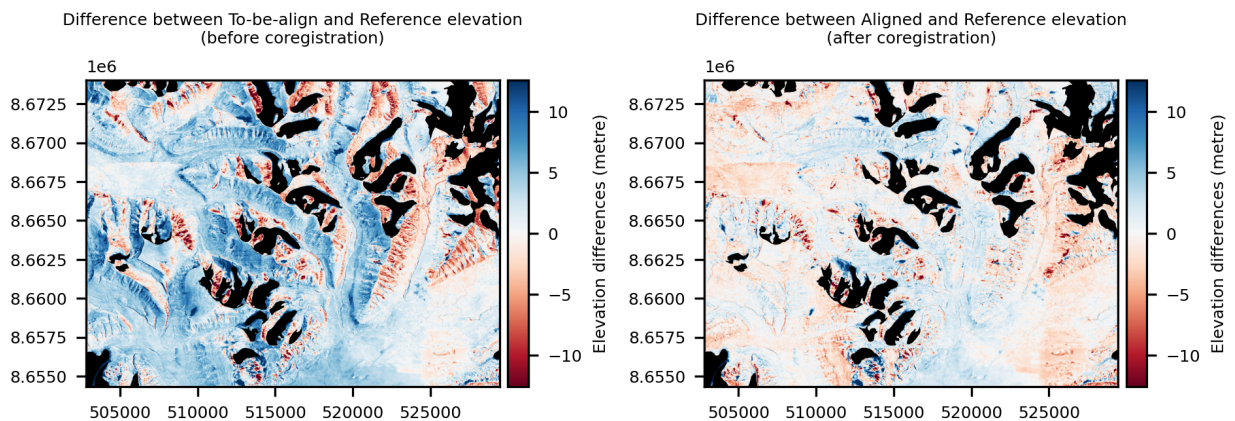
random      {'subsample_final': '10000'}

```

## Statistics

| Data        | Aligned elevation | Difference before coreg | Difference after coreg |
|-------------|-------------------|-------------------------|------------------------|
| Median      | 316.330           | 2.334                   | -0.048                 |
| NMAD        | 277.103           | 3.417                   | 2.050                  |
| Valid count | 1133607           | 1145221                 | 1133607                |

## Elevation differences



## Differences histogram

